

ESO208 COMPUTATIONAL METHODS IN ENGINEERING

ASSIGNMENT 7

MAXIMUM MARKS: 30

DUE DATE: 4 OCTOBER 2026

Please upload hand-written solutions on Gradescope. Typed solutions except codes and plots will not be graded. Please upload single PDF file for hand-written part.

Q.1. [15 marks] The following data is collected for a cross section of a river (y =distance from the bank, H = depth and U = velocity):

y (m)	0	2	4	6	8	10	12	14
H (m)	0	1	1.5	3	3.5	3.2	2	0
U (m/s)	0	0.1	0.12	0.2	0.25	0.3	0.15	0

The area of cross section A , discharge Q and average velocity $\bar{U}$ can be calculated as

$$A = \int_0^{y_{max}} H(y) dy$$

$$Q = \int_0^{y_{max}} H(y) U(y) dy$$

$$\bar{U} = \frac{Q}{A}$$

- (a) Write a code to calculate cross section area and discharge using the trapezoidal rule. [5]
- (b) Write a code to calculate the cross-section area and discharge using the Simpson's rule (use a combination of 1/3 and 3/8 rule). [7]
- (c) Make a table showing the comparison of Q , A and $\bar{U}$ calculated in (a) and (b). (Hand-written). [3]

Q.2. [15 marks] The amount of mass transported via a pipe over a period of time can be computed as

$$M = \int_{t_1}^{t_2} Q(t) c(t) dt$$

where M is mass in mg, t_1 = initial time (min), t_2 = final time (min), $Q(t)$ = flow rate (m^3/min) and $c(t)$ = concentration (mg/m^3). The mass transported between $t_1 = 2$ and $t_2 = 8$ min is to be determined for the following discharge and concentration functions

$$Q(t) = 8 + 5 \cos^2(0.4t)$$

$$c(t) = 5e^{-0.5t} + 3e^{0.15t}$$

(a) Write a code to calculate the amount of mass transported using Romberg integration with a tolerance of 0.1%. [10]

(b) Write the equations used in Romberg integration and the final output from your code (hand-written). [5]